Idea Assessment - 276 - Mohamed Abdurrahman Mohame

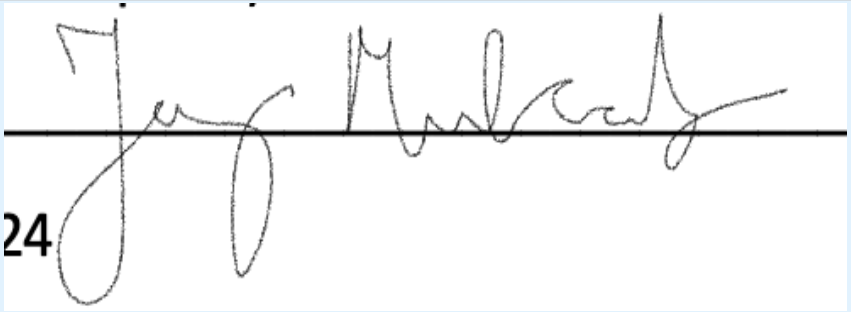
smartsheet

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1	Idea Assessment No.	276		
2	Project Title		Decision Advantage: AI-Enabled Hospital Flow Decision Support	
3	Date Submitted		07/09/26	
4	Name		**Mohamed Abdurrahman Mohamed**	
5	Role		Independent Innovator and Product Lead	
6	Organisation/Institution		Decision Advantage — independent innovation project	
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8	HIHI Contact Person		Ruan Kane	
9	**THE PROBLEM**			
10	Define your problem statement. Example: "A way to [address the problem] in [a specific population] in order to [achieve a desired outcome]." Example: "A way to reduce the incidence of diabetes in adults aged 40-60 in rural communities to improve health outcomes and reduce healthcare costs."		A way to help authorised acute-hospital flow teams recognise emerging pressure earlier and shorten the time from an operational signal to a coordinated, accountable response, so that available capacity is used more effectively and avoidable delays for patients are reduced.	
11	Identified Problem Briefly outline the problem / challenge / unmet need that your innovation / solution proposes to address.		Irish acute-hospital flow is constrained by beds, workforce, diagnostics and community capacity. These structural issues can be compounded by decision latency: avoidable time between a relevant operational signal becoming available, teams forming a shared view, an authorised decision being made and the agreed action being completed. Information on ED demand, admitted patients awaiting beds, occupancy, discharge status, delayed transfers and community capacity may sit across separate systems, reports and teams. Staff can spend time reconciling the operational picture, while emerging pressure may be recognised late and recovery actions may lack a clearly recorded owner or completion status. This can delay escalation, bed allocation, discharge coordination and activation of local recovery plans. Decision Advantage addresses this bounded coordination problem; it does not claim to replace the additional capacity, workforce or community services that the wider system requires.	
12	Extent of Problem Briefly outline the types of people who are affected by this problem. What groups of people are impacted by this problem How are they impacted?		The problem affects patients awaiting admission, inpatients whose next care step is delayed, people medically ready to leave hospital, and patients whose care depends on timely acute-community coordination. Older people and those with complex health or social-care needs may be particularly exposed to prolonged waits or time in an inappropriate care setting. It also affects patient-flow, bed-management, ED, discharge, acute-community and operational-leadership staff through repeated information gathering, hand-offs and escalation work. The HSE forecasts 1.74 million ED attendances in 2026, a 3.2% increase, and targets 85% of attendees being admitted or discharged within nine hours. It also requires 29 sites to operate centralised hubs with real-time information on ED and hospital flow, discharge and acute/community capacity, supported by site-specific trolley and delayed-transfer thresholds and recovery plans. These figures demonstrate the scale and strategic relevance of the operating environment; they do not, by themselves, establish how much delay is caused by decision latency. That contribution must be measured at a participating site.	
13	HIHI Opinion		The identified problem is well articulated. Similarly, the parties identified have been appropriately identified. As the innovator notes, the exact extent is difficult to quantify at present, but when further data becomes available this will be a key part of the case for this solution. The problem statement is framed at a fairly systemic level, with some of its language anticipating the shape of the proposed solution. It would be worth exploring the validity of this framing with frontline operational and clinical staff, to confirm it reflects how the problem is actually experienced, rather than relying solely on the innovator's own framing.	
14	**THE SOLUTION**			

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15	Proposed Solution Describe the innovation / solution. What is the solution? How does it address the identified problem?		Decision Advantage is a human-in-the-loop hospital operations intelligence platform built on the Integrated Predictive Care Intelligence (IPCI) framework. It supports one defined operational decision at a time through a simple five-step cycle: 1. Aggregate approved operational signals 2. Estimate near-term pressure 3. Compare against site-agreed thresholds 4. Present an evidence-linked, locally approved recovery action 5. Record acceptance, modification, rejection or escalation for audit and learning The prototype uses synthetic data and is strictly non-clinical. It does not diagnose, select treatment, prioritise patients, admit or discharge patients, allocate beds autonomously or execute actions without human approval. The platform addresses the identified problem by replacing reactive, fragmented decision-making with a predictable, auditable and standardised decision cycle that shortens decision latency and supports earlier coordinated action. - We propose a 12-week HIHI-supported evaluation pathway: - Co-design one or two operational decisions - Qualify data and compare forecasts/rules with simple baselines - Run outputs silently on prospective data - Test usability, comprehension and workflow fit - Make a documented “go, revise or stop” decision Controlled live use would only proceed once agreed safety, governance, data-quality and integration gates are met.	
16	Uniqueness Explain why you consider your innovation / solution is unique. What is different and unique about your solution? Can your solution's intellectual property (IP) be protected with a patent or by other means?		The innovation is not claimed to be unique because it uses dashboards or forecasting; established vendors already provide patient-flow, capacity and command-centre products. Its proposed differentiation is the way it structures a locally governed decision cycle for the Irish acute-hospital context: signal, forecast, local threshold, approved response, named owner, human decision, completion and learning. Evidence, data freshness, time horizon, uncertainty, overrides and an audit trail are intended to remain visible. The initial aggregate, read-only and non-diagnostic scope also provides a lower-risk route to testing value before patient-specific or autonomous functionality is considered. No patent filing has been made. Current protection is through founder-controlled source code and documentation, software copyright, project-specific architecture and configuration, interface designs, IPCI terminology, confidential know-how and controlled disclosure. Before commercial deployment, the project will complete an ownership and third-party-licence review, software bill of materials and contractual IP assignments. Trade-mark protection and patentability will be assessed professionally; patentability is not assumed because this field contains substantial prior art.	
17	Category of the solution		Support systems, such as electronic patient records, apps, e-solutions	
18	Stage of Development What stage of development has the solution reached?		Early prototyping and testing complete	
19	Intellectual Property Have the idea been protected?		No	
20	HIHI Opinion		The solution and 5 step cycle is well-explained, though could perhaps benefit from the inclusion of one or two examples/use cases to illustrate the process. A pilot study is described, though the need, framing and acceptability of the solution would need to be confirmed with clinicians and operational staff initially. The innovator demonstrates good knowledge of IP protection, though there may be concerns regarding how well the product could be protected on an ongoing basis in the absence of a patent. It would also be useful to understand how the innovator envisions securing access to the underlying operational data feeds the platform depends on (emergency department demand, bed occupancy, discharge status, etc.), given these typically sit in separate hospital systems with their own governance and integration requirements.	
21	**THE COMPETITION**			

22	Other Solutions Outline briefly any other solutions to this problem. What are the competitors to your solution, if they exist? Give name and description of other solutions on the market or in development		Global competitors include mature hospital operations and patient-flow platforms such as TeleTracking Operations IQ, LeanTaaS iQueue for Inpatient Flow, Qventus Inpatient Capacity and GE HealthCare Command Center. These systems combine real-time operational visibility with capacity forecasting, discharge workflows and predictive analytics. Electronic Health Record vendors also provide native alternatives. Epic Grand Central / Capacity Command Center supports real-time bed management, discharge planning and predictive analytics, while Oracle Health Patient Flow and Command Center offers near-real-time throughput, capacity and task-management functions. In Ireland, the most relevant comparator is the current operational workflow: HSE or locally developed flow hubs, EHR/bed-management modules, BI dashboards, spreadsheets, reports, huddles, telephone calls, email and messaging. For HIHI evaluation, Decision Advantage should therefore be compared not only with named vendor platforms but with the participating site's actual baseline process and planned digital investments.	
23	Advantages of the proposed Solution Explain why you consider the proposed solution has advantages over existing research/competitors that you are aware of. How is the proposed solution better than other solutions?		Decision Advantage's potential advantages are design hypotheses to be tested during evaluation, not established performance claims. Compared with existing research and enterprise platforms, the solution offers several early-stage advantages: 1. Accountable Execution: Each prompt is linked to evidence, a local threshold, an approved action, a human decision, a named owner and a tracked completion status, supporting consistent and auditable operational decisions. 2. Localised Flexibility: The platform is configurable to individual HSE site definitions and recovery plans rather than imposing a generic vendor workflow. 3. Reduced Risk Surface: An aggregate-first, read-only and non-diagnostic architecture minimises initial GDPR, integration and clinical-risk barriers. 4. Rigorous Baseline Validation: The staged evaluation pathway compares performance against simple baselines and uses silent prospective and human-factors testing before influencing live operations. 5. Scalable Structure: Reusable workflow and metric components support future multi-site scaling while preserving local thresholds, roles and adapters. These hypotheses will be tested against existing hospital systems. Progression will stop or be redesigned if data cannot support the selected decision, users cannot interpret or reject outputs safely, alert burden becomes unacceptable or incremental value over current practice is not demonstrated.	
24	HIHI Opinion		The competitor landscape has been researched thoroughly and the innovator shows a good understanding of the market. However, this is a busy and well-established space, with mature specialist vendors, EHR-based solutions and existing local hospital systems already offering overlapping functionality. The proposed differentiation currently rests largely on configuring the solution to local HSE workflows and providing an auditable decision cycle. This alone does not yet demonstrate a sufficiently strong competitive advantage. The innovator should identify a specific gap that existing or planned solutions do not adequately address and demonstrate clear additional value.	
25	**THE MARKET**			

26	Potential Customers Outline who you consider the customers for your solution will be. Who will use the proposed solution? Who would purchase the proposed solution when it is market-ready?		End users — who will use the solution day-to-day - ED shift leads, flow coordinators and nurse managers - Bed managers and patient-flow teams - Discharge coordinators and acute-community transition teams - Hospital operations and performance managers - Regional urgent and emergency care operational staff Implementation and assurance stakeholders — required for adoption - Digital/ICT and integration teams - Data protection and information governance - Cybersecurity - Clinical safety and clinical governance - Data science and analytics Economic buyers — who would purchase the solution - Acute hospital executives or operations directors - HSE Health Regions - National HSE programmes, depending on procurement route Future buyers (post-validation) - Private hospital groups in Ireland - UK NHS trusts and integrated care systems - Selected European hospital systems Commercial model - Annual hospital- or regional-level subscription - Implementation, integration, validation, training and support priced separately - Purchase decisions will depend on reference-site evidence of workflow value, integration effort, safety and total cost Priority initial customer: A high-pressure acute hospital with engaged operational leadership and clear HSE data-governance support.	
27	Potential Market Size Include any market research you have conducted to date. What is the size of the market for the proposed solution in Ireland? What is the size of the market for the proposed solution internationally? What would the price range for the proposed solution be, when it is market ready ?		Ireland: A bottom-up estimate based on hospital sites is used. OECD data identify 56 general hospitals in Ireland (2024). Applying €75,000–€250,000 per site gives an indicative Irish TAM of €4.2m–€14.0m, with a base case of €8.4m. The HSE National Service Plan 2026 identifies 29 sites for real-time flow hubs, thresholds and recovery plans. These form a focused initial SAM of €2.175m–€7.250m, with a base case of €4.35m. These are planning scenarios, not sales forecasts. Regional or national procurement may result in fewer, larger contracts. International: Comparable OECD general-hospital counts: - UK: 807 (2022) - Finland: 104 (2023) - Netherlands: 77 (2023) Total: 988 sites. Applying the same licence range gives an upper-bound TAM of €74.1m–€247.0m annually. This is a high-level TAM indicator. It requires filtering for procurement structure, digital maturity, incumbent systems, regulation and support capacity. Denmark, Sweden and Norway remain target markets but are excluded due to unavailable comparable OECD counts. Price range when market-ready - Software licence: €75,000–€250,000 per hospital site per year - Base case: €150,000 - Additional costs: implementation, integration, validation, training and support Likely model: annual hospital- or regional-level subscription; national/regional contracts may reduce per-site fees	

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28	HIHI Opinion		Customer segmentation and market sizing approach is both appropriate and transparent. The innovator is realistic in framing these as 'planning scenarios' rather than forecasts. Good awareness of the buying logic and stakeholders involved. The estimated license price figures are acknowledged to be working assumptions, to be refined via customer discovery, procurement testing, implementation-cost analysis and an independently verified site value case. They are also given without reference to the prices of competitor offerings. Benchmarking against these would potentially strengthen the economic case. The accompanying commercial risks and mitigations table is a notable addition, demonstrating a realistic grasp of commercial risk.	
29	**ATTACHMENT**			
30	Document attached		documents attached	
31	**NEXT STEPS**			
32	Summary and proposed Actions for Innovator		Problem: Well-articulated and appropriately scoped coordination problem. As the applicant acknowledges the full extent of the problem would still need to be quantified. Solution: Clearly described five-step cycle, well-scoped as a human-in-the-loop, non-diagnostic, read-only platform, but still at an early stage with no defined use case value. Market: Thorough and transparent. Benchmarking required against existing competitor pricing Competition: A busy market. Well-researched generally. Differentiation claim requires further work, ideally through benchmarking against competitor product capacity for integration within current HSE/hospital settings. Note, this may require going beyond existing marketing materials. Next Steps: a. Speak with the relevant HSE Technology and Transformation and patient-flow teams to understand whether there is a real gap for this solution within the HSE. b. Focus on one clear use case and show what your solution would do better or differently from systems already available or planned. c. Explore how you would access and integrate the required hospital data, and what a realistic route to adoption would look like	
33	Proposed Actions for HIHI		a. Signpost the innovator to the relevant HSE Technology and Transformation and patient-flow teams to explore whether there is a clear need and fit for the solution. b. No further HIHI evaluation or support is recommended at this stage. Consider re-engagement if the innovator can demonstrate a clear gap and added value over existing or planned solutions.	
34	Prepared by HIHI		Ruán Kane	
35	Signed by HIHI Director			
36	Date		05.10.26	
37	HIHI Director's Comment			
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